

electronic system called moving map display, which displays the map of the terrain over which the pilot flies. As the pilot flies ahead, the map also scrolls up accordingly. Waypoints are updated in the display through this datalink. In the carrier, there are 62 Link-11 datalink channels available. Each datalink is capable of communicating as far as 300km.

Link-16. Link-16 is a node-less, jam-resistant, high-speed, encrypted digital datalink. It operates in 960MHz-1215MHz band. Based on TDMA technique, it is used by aircraft, ships and even ground forces to exchange their tactical pictures in near-real-time.

Typically, commanders of the CSG at sea can even get tactical pictures around ground forces operating deep inside enemy territory. Link-16 supports exchange of text messages, imagery data and provides two channels for digital voice communications with a data rate of 2.4kbps and/or 16kbps in any combination. The range is around 500km and there are 255 Link-16 datalink channels available in the carrier.

Officer-in-tactical command information exchange sub-system (OTCIX). This is a two-way UHF satcom based datalink system for intra- and inter-CSG communication. This link supports the exchange of targeting information for multi-targeting of sub-surface, surface and aerial targets. A ship can get radar track data from other ships through this link even while not using its radar at all. This is called over-the-horizon (OTH) targeting.

Tactical data information exchange sub-system (TADIXS) downlink. This is a shore-to-ship one-way satellite datalink for the ships to receive mission plans for launching Tomahawk Land Attack

Missiles (TLAMs).

Sonobuoy datalink. This datalink connects sonobuoys with the under-sea warfare system present in destroyers. These dispensable sonobuoys are dropped by the submarine hunting helicopters in the areas suspected with enemy submarine activity. The under-sea warfare system deduces the submarine scenario through these datalinks.

C² systems

In a carrier, the storey above the flight-deck is called flag bridge. Commander of the CSG controls all operations from a command centre called tactical flag command centre

Similarly, captain of the carrier has another parallel ACDS display at his command centre called combat direction centre (CDS) from where he or she gets the tactical picture around the carrier.

Global C² systems-maritime (GCCS-M). GCCS-M is a system that receives, processes, displays and manages data on the readiness of neutral, friendly and hostile forces in near-real-time. It does this via external communication channels, local area networks and direct interfaces with other systems. Any information the CSG commander required for the mission, be it intelligence or meteorological data, is sent

through this system from the higher ups. It is through this GCCS-M system that the commander of the CSG takes orders from the higher command and passes situation reports to the higher command.

For a CSG commander, GCCS-M aids in the decision-making process by displaying relevant tactical information. For example, the CSG is anchored 300km from enemy shore. Job of the CSG is to neutralise an enemy nuclear missile base situated 200km

inland. The condition for neutralisation is that the missile base readies itself for a nuclear missile launch. But, how will the CSG commander know that the nuclear missile base is ready for missile launch?

This information can only be gathered by people of the likes of a man drinking a Vodka Martini and who is addressed as 007. Purely, the data he collects is intelligence related. Such information is not available through the CSG's resources for the CSG commander. So in this case, intelligence has to be supplied by the high command to the CSG commander.

If the enemy nuclear missile base readies for a missile launch, the



Fig. 31: A GCCS-M display (Image courtesy: US Navy)

(TFCC), situated in this bridge. In naval parlance, flag indicates something related to the commander of a fleet.

Commander of the CSG gets instructions from his high command through a satellite based C² system called global command and control systems-maritime (GCCS-M). Through this system, commander of the CSG provides situation reports to his high command.

A system called advanced combat direction system (ACDS) creates the tactical picture around the CSG. Display of the ACDS gives the CSG commander the tactical picture around the CSG for taking combat decisions.